[redacted] CHE 422 4/10/2020

# Biochemical Engineering Project #1

10/10 ☆☆☆

## Description and Function of Target Protein

The protein was identified as a **mitotic deacetylase-associated SANT domain protein (MIDEAS)** by using the BLAST tools on the UniProt website (See figure 1) [1]. This specific protein originates from the *Homo sapiens* (Humans). The MIDEAS protein is thought to have "DNA-binding transcription factor activity; transcription factor binding activity; and transcription regulatory region DNA binding activity" [2]. It is also thought to be involved in the regulation of DNA transcription. It focuses on the histone deacetylase complex and transcription factor complex [2].

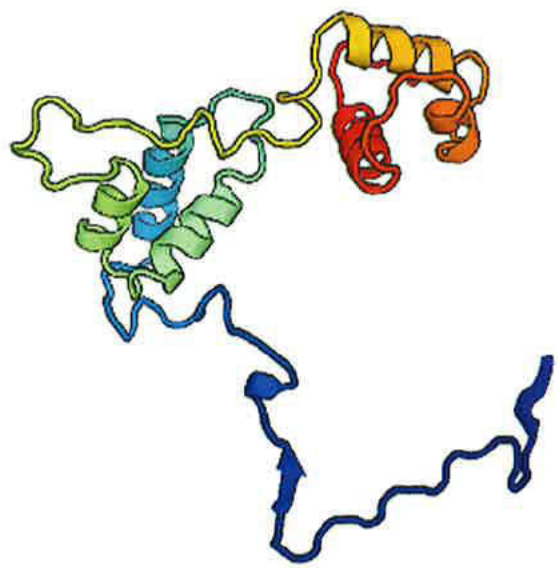

**Figure 1:** The three-dimensional structure of mitotic deacetylase-associated SANT domain protein (MIDEAS). MIDEAS is thought to be involved in the regulation of DNA transcription [2]. Figure adapted from UnitProt [1].

## DNA Sequence of the Protein

*Pichia pastoris* was chosen as the organism to produce our protein. The following table gives the codon bias for *Pichia pastoris* when reverse translating the protein sequence into a DNA sequence [3].

| AA | Codon | Number | /1000 | Fraction | AA | Codon | Number | /1000 | Fraction |
|---|---|---|---|---|---|---|---|---|---|
| Gly | GGG | 50527.00 | 11.12 | 0.15 | Thr | ACG | 63696.00 | 14.02 | 0.26 |
| Gly | GGA | 39036.00 | 8.59 | 0.12 | Thr | ACA | 35995.00 | 7.93 | 0.15 |
| Gly | GGT | 114185.00 | 25.14 | 0.34 | Thr | ACT | 43256.00 | 9.52 | 0.18 |
| Gly | GGC | 130043.00 | 28.63 | 0.39 | Thr | ACC | 103121.00 | 22.70 | 0.42 |
| Glu | GAG | 83804.00 | 18.45 | 0.32 | Trp | TGG | 65630.00 | 14.45 | 1.00 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Glu | GAA | 179460.00 | 39.51 | 0.68 | End | TGA | 4428.00 | 0.97 | 0.30 |
| Asp | GAT | 146794.00 | 32.32 | 0.63 | Cys | TGT | 23461.00 | 5.17 | 0.45 |
| Asp | GAC | 87759.00 | 19.32 | 0.37 | Cys | TGC | 28747.00 | 6.33 | 0.55 |
| Val | GTG | 115687.00 | 25.47 | 0.36 | End | TAG | 1172.00 | 0.26 | 0.08 |
| Val | GTA | 51020.00 | 11.23 | 0.16 | End | TAA | 9006.00 | 1.98 | 0.62 |
| Val | GTT | 86572.00 | 19.06 | 0.27 | Tyr | TAT | 75774.00 | 16.68 | 0.58 |
| Val | GTC | 67356.00 | 14.83 | 0.21 | Tyr | TAC | 55847.00 | 12.30 | 0.42 |
| Ala | GCG | 146264.00 | 32.20 | 0.34 | Leu | TTG | 60322.00 | 13.28 | 0.13 |
| Ala | GCA | 93390.00 | 20.56 | 0.22 | Leu | TTA | 62823.00 | 13.83 | 0.13 |
| Ala | GCT | 73677.00 | 16.22 | 0.17 | Phe | TTT | 100128.00 | 22.05 | 0.57 |
| Ala | GCC | 113412.00 | 24.97 | 0.27 | Phe | TTC | 74885.00 | 16.49 | 0.43 |
| Arg | AGG | 7423.00 | 1.63 | 0.03 | Ser | TCG | 39546.00 | 8.71 | 0.15 |
| Arg | AGA | 12345.00 | 2.72 | 0.05 | Ser | TCA | 35837.00 | 7.89 | 0.13 |
| Ser | AGT | 41544.00 | 9.15 | 0.15 | Ser | TCT | 42367.00 | 9.33 | 0.16 |
| Ser | AGC | 70867.00 | 15.60 | 0.26 | Ser | TCC | 40365.00 | 8.89 | 0.15 |
| Lys | AAG | 51685.00 | 11.38 | 0.25 | Arg | CGG | 25751.00 | 5.67 | 0.10 |
| Lys | AAA | 156169.00 | 34.38 | 0.75 | Arg | CGA | 16607.00 | 3.66 | 0.07 |
| Asn | AAT | 84846.00 | 18.68 | 0.46 | Arg | CGT | 93997.00 | 20.70 | 0.37 |
| Asn | AAC | 98018.00 | 21.58 | 0.54 | Arg | CGC | 96053.00 | 21.15 | 0.38 |
| Met | ATG | 123604.00 | 27.21 | 1.00 | Gln | CAG | 130898.00 | 28.82 | 0.66 |
| Ile | ATA | 24233.00 | 5.34 | 0.09 | Gln | CAA | 67129.00 | 14.78 | 0.34 |
| Ile | ATT | 135873.00 | 29.92 | 0.50 | His | CAT | 57585.00 | 12.68 | 0.57 |
| Ile | ATC | 111878.00 | 24.63 | 0.41 | His | CAC | 43743.00 | 9.63 | 0.43 |
| Pro | CCG | 101467.00 | 22.34 | 0.51 | Leu | CTG | 231373.00 | 50.94 | 0.49 |
| Pro | CCA | 38663.00 | 8.51 | 0.20 | Leu | CTA | 18067.00 | 3.98 | 0.04 |
| Pro | CCT | 32678.00 | 7.19 | 0.17 | Leu | CTT | 51442.00 | 11.33 | 0.11 |
| Pro | CCC | 24383.00 | 5.37 | 0.12 | Leu | CTC | 48147.00 | 10.60 | 0.10 |

The DNA sequence was determined by imputing the one letter protein sequence into an online reverse translator (*www.bioinformatics.org/sms2/rev_trans.html*) [4]. The translator then generated the following DNA base sequence.

```
AGGCCTCATCATCATCATCATCATGACGATGACGATAAG
ATGAACCTGCAGGCGCAGCCGAAAGCGCAGAACAAACGCAAACGCTGCCTGTTTGGCGGC
CAGGAACCGGCGCCGAAAGAACAGCCGCCGCCGCTGCAGCCGCCGCAGCAGAGCATTCGC
GTGAAAGAAGAACAGTATCTGGGCCATGAAGGCCCGGGCGGCGCGGTGAGCACCAGCCAG
CCGGTGGAACTGCCGCCGCCGAGCAGCCTGGCGCTGCTGAACAGCGTGGTGTATGGCCCG
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GAACGCACCAGCGCGGCGATGCTGAGCCAGCAGGTGGCGAGCGTGAAATGGCCGAACAGC
GTGATGGCGCCGGGCCGCGGCCCGGAACGCGGCGGCGGCGGCGTGAGCGATAGCAGC
TGGCAGCAGCAGCCGGGCCAGCCGCCGCCGCATAGCACCTGGAACTGCCATAGCCTGAGC
CTGTATAGCGCGACCAAAGGCAGCCCGCATCCGGGCGTGGGCGTGCCGACCTATTATAAC
CATCCGGAAGCGCTGAAACGCGAAAAAGCGGGCGGCCCGCAGCTGGATCGCTATGTGCGC
CCGATGATGCCGCAGAAAGTGCAGCTGGAAGTGGGCCGCCCGCAGGCGCCGCTGAACAGC
TTTCATGCGGCGAAAAAACCGCCGAACCAGAGCCTGCCGCTGCAGCCGTTTCAGCTGGCG
TTTGGCCATCAGGTGAACCGCCAGGTGTTTCGCCAGGGCCCGCCGCCGCCGAACCCGGTG
GCGGCGTTTCCGCCGCAGAAACAGCAGCAGCAGCAGCAGCCGCAGCAGCAGCAGCAG
CAGCAGGCGGCGCTGCCGCAGATGCCGCTGTTTGAAAACTTTTATAGCATGCCGCAGCAG
CCGAGCCAGCAGCCGCAGGATTTTGGCCTGCAGCCGGCGGGCCCGCTGGGCCAGAGCCAT
CTGGCGCATCATAGCATGGCGCCGTATCCGTTTCCGCCGAACCCGGATATGAACCCGGAA
CTGCGCAAAGCGCTGCTGCAGGATAGCGCGCCGCAGCCGGCGCTGCCGCAGGTGCAGATT
CCGTTTCCGCGCCGCAGCCGCCGCCTGAGCAAAGAAGGCATTCTGCCGCCGAGCGCGCTG
GATGGCGCGGGCACCCAGCCGGGCCAGGAAGCGACCGGCAACCTGTTTCTGCATCATTGG
CCGCTGCAGCAGCCGCCGCCGGGCAGCCTGGGCCAGCCGCATCCGGAAGCGCTGGGCTTT
CCGCTGGAACTGCGCGAAAGCCAGCTGCTGCCGGATGGCGAACGCCTGGCGCCGAACGGC
CGCGAACGCGAAGCGCCGGCGATGGGCAGCGAAGAAGGCATGCGCGCGGTGAGCACCGGC
GATTGCGGCCAGGTGCTGCGCGGCGGCGTGATTCAGAGCACCCGCCGCCGCCGCCGCGCG
AGCCAGGAAGCGAACCTGCTGACCCTGGCGCAGAAAGCGGTGGAACTGGCGAGCCTGCAG
AACGCGAAAGATGGCAGCGGCAGCGAAGAAAAACGCAAAAGCGTGCTGGCGAGCACCACC
AAATGCGGCGTGGAATTTAGCGAACCGAGCCTGGCGACCAAACGCGCGCGCGAAGATAGC
GGCATGGTGCCGCTGATTATTCCGGTGAGCGTGCCGGTGCGCACCGTGGATCCGACCGAA
GCGGCGCAGGCGGGCGGCCTGGATGAAGATGGCAAAGGCCCGGAACAGAACCCGGCGGAA
CATAAACCGAGCGTGATTGTGACCCGCCGCCGCAGCACCCGCATTCCGGGCACCGATGCG
CAGGCGCAGGCGGAAGATATGAACGTGAAACTGGAAGGCGAACCGAGCGTGCGCAAACCG
AAACAGCGCCCGCGCCCGGAACCGCTGATTATTCCGACCAAAGCGGGCACCTTTATTGCG
CCGCCGGTGTATAGCAACATTACCCCGTATCAGAGCCATCTGCGCAGCCCGGTGCGCCTG
GCGGATCATCCGAGCGAACGCAGCTTTGAACTGCCGCCGTATACCCCGCCGCCGATTCTG
AGCCCGGTGCGCGAAGGCAGCGGCCTGTATTTTAACGCGATTATTAGCACCAGCACCATT
CCGGCGCCGCCGCCGATTACCCCGAAAAGCGCGCATCGCACCCTGCTGCGCACCAACAGC
GCGGAAGTGACCCCGCCGGTGCTGAGCGTGATGGGCGAAGCGACCCCGGTGAGCATTGAA
CCGCGCATTAACGTGGGCAGCCGCTTTCAGGCGGAAATTCCGCTGATGCGCGATCGCGCG
CTGGCGGCGGCGGATCCGCATAAAGCGGATCTGGTGTGGCAGCCGTGGGAAGATCTGGAA
AGCAGCCGCGAAAAACAGCGCCAGGTGGAAGATCTGCTGACCGCGGCGTGCAGCAGCATT
TTTCCGGGCGCGGGCACCAACCAGGAACTGGCGCTGCATTGCCTGCATGAAAGCCGCGGC
GATATTCTGGAAACCCTGAACAAACTGCTGCTGAAAAAACCGCTGCGCCCGCATAACCAT
CCGCTGGCGACCTATCATTATACCGGCAGCGATCAGTGGAAAATGGCGGAACGCAAACTG
TTTAACAAAGGCATTGCGATTTATAAAAAAGATTTTTTTCTGGTGCAGAAACTGATTCAG
ACCAAAACCGTGGCGCAGTGCGTGGAATTTTATTATACCTATAAAAAACAGGTGAAAATT
GGCCGCAACGGCACCCTGACCTTTGGCGATGTGGATACCAGCGATGAAAAAAGCGCGCAG
GAAGAAGTGGAAGTGGATATTAAAACCAGCCAGAAATTTCCGCGCGTGCCGCTGCCGCGC
CGCGAAAGCCCGAGCGAAGAACGCCTGGAACCGAAACGCGAAGTGAAAGAACCGCGCAAA
GAAGGCGAAGAAGAAGTGCCGGAAATTCAGGAAAAAGAAGAACAGGAAGAAGGCCGCGAA
CGCAGCCGCCGCGCGGCGGCGGTGAAAGCGACCCAGACCCTGCAGGCGAACGAAAGCGCG
AGCGATATTCTGATTCTGCGCAGCCATGAAAGCAACGCGCCGGGCAGCGCGGGCGGCCAG
GCGAGCGAAAAACCGCGCGAAGGCACCGGCAAAAGCCGCCGCGCGCTGCCGTTTAGCGAA
AAAAAAAAAAAAACCGAAACCTTTAGCAAAACCCAGAACCAGGAAAACACCTTTCCGTGC
AAAAAATGCGGCCGC**GTCGAC**

**NOTE:**

AGGCCT: Is the restriction site that StuI recognizes. It cleaves the site at AGG | CCT and creates blunt ends.

GTCGAC: Is the restriction site that SalI recognizes. It cleaves the site at G | TCGAC and creates sticky ends.

CATCATCATCATCATCAT: Is the polyhistidine tag (x6) or the HisTag incorporated into the MIDEAS gene for purification steps.

GACGATGACGATAAG: Is the EK recognition site. This is the site that the EK protease recognizes to cleave the protein.

**Considerations for Host Cell and Expression Conditions**

The most common host cells for protein expression are mammalian, insect, yeast, and bacterial cells. A table of the most common advantages and disadvantages is described below.

**Table 1:** Comparison between the mammalian, insect, yeast, and bacterial cells. The information in this table is directly taken from *ThermoFisher Scientific* [5].

| ***Cell Types*** | ***Advantages*** | ***Disadvantages*** |
|---|---|---|
| **Mammalian** | • Highest level of protein processing<br>• Can produce proteins either transiently, or by stable expression<br>• Robust optimized transient systems for rapid, ultrahigh-yield protein production | • g/L yields only possible in suspension cultures<br>• More demanding culture conditions |
| **Insect** | • Similar to mammalian protein processing<br>• Can be used in static or suspension cell culture | • More demanding culture conditions than prokaryotic systems<br>• Production of recombinant baculovirus vectors is time consuming |
| **Yeast** | • Eukaryotic protein processing<br>• Scalable up to fermentation (g/L)<br>• Simple media requirements | • Fermentation required for high yields<br>• Growth conditions may require optimization |
| **Bacterial** | • Scalable<br>• Low cost<br>• Simple culture conditions | • Protein solubility<br>• Difficult to express proteins > 50 kDA<br>• Cannot perform post-translational modification |

The molecular weight of the MIDEAS protein is 115 kDa. The MIDEAS protein also requires post-translational modification like the ubiquitination at Lys843 and modification sites at PhosphoSitePlus [6]. A bacterial host cell therefore cannot be used to express the protein as the protein is too large and requires post-translational modification. The yeast cells require the simplest media requirements, lowest costs, easiest scalabilities, and the ability to express our protein. Therefore, in the expression of the MIDEAS protein, the *Pichia* Expression system by Invitrogen™ that uses the *Pichia pastoris* yeast cells will be used [7]. The *Pichia* Expression system is claimed to be as easy to use as bacterial systems and is scalable up to 10,000 L batch fermenters.

The *Pichia* Expression System uses the methylotrophic yeast *Pichia pastoris* for high-level recombinant protein expression [7]. *P. pastoris* will use methanol as a carbon source in the absence of glucose. The promoter that controls the gene expression of the desired gene is the alcohol oxidase (AOX1). AOX1 catalyzes the first step in the methanol metabolism of the yeast. The following fermentation conditions are required for the optimal growth of the yeast and the optimal expression of the protein [7].

- Temperature of 30°C
- Dissolved oxygen of >20%
- A pH of 5.0-6.0
- An agitation speed of 500-1500 rpm

**Expression Vector and Cloning Site Information**

The chosen expression vector is the pHIL-S1 expression vector. This vector allows the secreted expression of the protein. The expression vector is diagramed below:

**Comments for pHIL-S1:**
**8260 nucleotides**

5' *AOX1* promoter fragment: bases 1-941
5' *AOX1* primer site: bases 856-876
PHO1 secretion signal (S): bases 942-1007
Multiple Cloning Site Region: bases 1006-1026
3' *AOX1* primer site: bases 1099-1119
3' *AOX1* transcription
termination (TT) fragment: bases 1025-1190
*HIS4* ORF: bases 4286-1753
3' *AOX1* fragment: bases 4641-5397
pBR322 origin: bases 6556-5937
f1 origin of replication: bases 6651-7106
Ampicillin resistance gene: bases 7922-7062

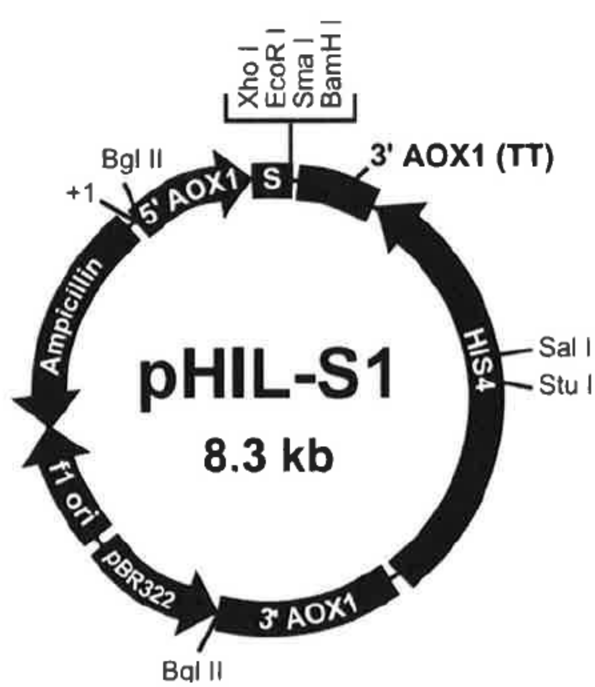

**Figure 2:** This map shows the size of the expression vector and the location of the various features of the vector. Figure adapted from *ThermoFisher Scientific* [7].

The information about the multiple cloning site region is given below.

```
773                                                          AOX1 mRNA 5' end (825)
ACAGGCAATA TATAAACAGA AGGAAGCTGC CCTGTCTTAA ACCTTTTTTT TTATCATCAT

                               5' AOX1 primer site (856-876)
TATTAGCTTA CTTTCATAAT TGCGACTGGT TCCAATTGAC AAGCTTTTGA TTTTAACGAC

                                                      PHO1(942-1007)
TTTTAACGAC AACTTGAGAA GATCAAAAAA CAACTAATTA TTCGAAACG ATG TTC TCT
                                                      Met Phe Ser

CCA ATT TTG TCC TTG GAA ATT ATT TTA GCT TTG GCT ACT TTG CAA TCT GTC
Pro Ile Leu Ser Leu Glu Ile Ile Leu Ala Leu Ala Thr Leu Gln Ser Val

 PHO1 cleavage site
 Xho I*      EcoR I  Sma I  BamH I
TTC GCT CGA GAA TTC CCC GGG ATC CTT AGA CAT GAC TGT TCC TCA GTT CAA
Phe Ala Arg Glu Phe Pro Gly Ile Leu Arg His Asp Cys Ser Ser Val Gln

                                       Stop (1083)
GTT GGG CAC TTA CGA GAA GAC CGG TCT TGC TAG ATTCTAATCA AGAGGATGTC
Val Gly His Leu Arg Glu Asp Arg Ser Cys ***

3' AOX1 primer site (1099-1119)
AGAATGCCAT TTGCCTGAGA GATGCAGGCT TCATTTTTGA TACTTTTTTA TTTGTAACCT

           AOX1 mRNA 3' end (1190)
ATATAGTATA GGATTTTTTT TGTCA
```

* If the *Xho* I site (which is part of the *PHO1* signal cleavage sequence) is used for cloning, it must be recreated in order for efficient cleavage of the fusion protein to occur.

**Figure 3:** This is the sequence and details of the multiple cloning site region in the pHIL-S1 expression vector. Figure adapted from *ThermoFisher Scientific* [7].

The insertion of our desired gene will be in the HIS4 region of the vector. The HIS4 region codes for the histidinol dehydrogenase gene which allows the *Pichia* host to synthesize histidine. Without the gene, the *Pichia* host cannot synthesize histidine and therefore cannot grow on histidine-deficient medium. This allows for the efficient screening of the yeast that have the transformed vector into the DNA as it will disrupt the HIS4 gene [7]. The restriction sites on the HIS4 gene are Sal I and Stu I.

**Molecular Cloning Procedures**

**Step 1. Isolate the Desired DNA Sequence from the Organism**

a. Isolate the gene of interest from the organism.
b. For mass production of the protein, a eukaryotic cell is chosen to clone/express the protein. The eukaryotic cell chosen will be a *Pichia pastoris* yeast cells.

**Step 2. Amplify the DNA sequence and add restriction sites by performing PCR**

a. Denature the isolated DNA strands from your organism at 94 °C for 1 minute.

b. Then anneal your DNA strand at 54 °C for 45 seconds by adding in your forward and reverse primers for the gene you want to amplify.
    i. The forward and reverse primers required for the MIDEAS DNA sequence to insert the StuI restriction site are AGGCCTCATCATCATCATCATCATGACGATGACGATAAGATGAACCTGC and TCCGGAGTAGTAGTAGTAGTAGTACTGCTACTGCTATTCTACTTGGACG.
    ii. The forward and reverse primers required for the MIDEAS DNA sequence to insert the SalI restriction site are ATGCGGCCGCGTCGAC and TACGCCGGCGCAGCTG.
c. Extend your partially completed DNA strand by adding in DNA Taq polymerase and dNTPs at a temperature of 72 °C for 2 minutes.
d. Repeat steps a-c 30 to 40 times.
e. This will amplify your target gene and incorporate the SalI and StuI endonuclease cutting sites.

**Step 3. Insert the DNA sequence into the pHIL-S1 Expression Vector**

a. Use restriction enzyme StuI and SalI to cut the pHIL-S1 plasmid at the HIS4 site region. This creates a linear DNA fragment with one sticky end and one blunt end. Two rustication sites are used to control the direction of the inserted DNA fragment.
b. Use restriction enzyme StuI and SalI to cut the desired DNA fragment at the respective restriction cutting sites. This creates one sticky end and one blunt end on the DNA fragment.
c. Add T4 DNA ligase and a buffer solution containing ATP to the fragment in order to connect the target DNA fragment to the pHIL-S1 vector, creating a circular plasmid.
d. Add the endonuclease BgI II to cut the plasmid twice to separate the gene of interest and the bacterial genes (i.g. Ampicillin resistant genes and site of origin). This creates two linear strands of DNA.

**Step 4. Transform the pHIL-S1 Expression Vector into Yeast Cells**

a. Mix the *Pichia pastoris* cells with the linearized DNA.
b. Incubate the cells on ice for 5 minutes.
c. Pulse the cells with electricity to cause sufficient electroporation.
d. Then add ice cold sorbitol.
e. Then transfer cells to an agar plate.

**Step 5. Perform a Screen on the Yeast Cells.**

a. Transfer the yeast cells to an agar dish with methanol as the only carbon source and that is histidine deficient. This will prevent the growth of the yeast cells that do not have the AOXI gene and prevent the growth of yeast cells that have the HIS4 gene disrupted.

b. Extract the small colonies that show now growth for further isolation and confirmation of the inserted genes. Colonies that survived but did not grow confirm that the yeast cells have the AOXI gene and a dysfunctional HIS4 gene.

**Step 6. Isolate and Digest the DNA**

a. Transfer the isolated colonies to a new agar dishes for further growth.
b. Use detergents to lyse some of the yeast that are part of the isolated colony from step 7 to release the cell's DNA.
c. Use a centrifuge to separate out the DNA from other cell debris.
d. Use the StuI and SalI restriction enzyme to cut the DNA into fragments.

**Step 7. Use Gel electrophoresis For Gene Verification**

a. Perform an agarose gel electrophoresis on the digested DNA sample.
    i. Load the DNA sample into one of the lanes in the gel.
    ii. Load a base pair standard sample into another lane in the gel.
    iii. Run the gel electrophoresis device for an appropriate amount of time depending on the wt% of the agarose gel and voltage used.
b. Stain the gel with ethidium bromide.
c. Use a UV light to see the DNA bands in the gel and take a picture of the gel with the glowing DNA bands.
d. Use the standard DNA bands to determine the size the DNA fragments from the plasmid.
e. Once the size of the DNA fragments is known, they can be compared to the expected size of the fragments if the gene was properly inserted into the plasmid.
    i. If the target gene was inserted into the yeast DNA, three fragments should be present in the gel electrophoresis. One fragment should appear at 3135 bp (inserted gene) and the other two fragments should appear much bigger (rest of yeast DNA).
    ii. If the target gene was not inserted into the plasmid, one fragments should be present in the gel electrophoresis. Two fragments should appear as it is the rest of the DNA of the yeast.
f. Perform Dideoxy DNA sequencing of the isolated gene.
    i. The forward and reverse primers required for the Dideoxy DNA sequencing are AGGCCTCATCATCATCATCATCATGACGATGACGATAAG and GTCGAC. This ensures that only the target gene is being sequenced.
g. Perform the next steps with the colonies that have the confirmed inserted and correctly sequenced gene. There is no need to check the orientation of the gene as

there is only one way the gene could have been inserted because there was one blunt and sticky end.

**Step 8. Start producing the protein**

a. Transfer the yeast colony that has the inserted desired gene to a growth media that has no methanol in it. This causes the inhibition of the transcription of the target gene while the yeast cells grow.
b. Once the yeast cells have a large enough concentration in your growth media to produce a sufficient amount of desired MIDEAS protein, add methanol to induce the transcription of the desired gene. Once induced, the cells will secrete the desired proteins.

**Step 9. Isolate the Desired Protein**

a. Since the cells secrete the desired proteins, take some of the media for further analysis.

**Step 10. Confirm the Protein Weight and Fidelity**

a. Denature the proteins (the media) using BME and heat. Add SDS to the denatured proteins.
b. Run a Polyacrylamide Gel Electrophoresis (PAGE) with the proteins along with a standard MIDEAS protein.
c. Add Coomassie Brilliant Blue to the gel to stain the protein blue.
d. Confirm that the molecular weight of isolated protein matches the molecular weight of the standard MIDEAS protein.
e. Transfer the proteins to a nitrocellulose blotting paper.
f. Incubate the blot with a generic protein, such as a milk protein, which bind to any remaining sticky places on the nitrocellulose.
g. The primary antibody will be conjugated with another antibody with the enzyme alkaline phosphatase. The primary antibody will attach to the one of the epitopes presents present on the MIDEAS protein.
h. The substrate *Fast Red* will be added to detect the antibody and the produced protein.
i. If the dye indicates the presence of the protein and the molecular weight of that isolated protein matches the standard MIDEAS protein weight, we have successfully isolated our protein.

**Step 11. Production of MIDEAS Protein Through Batch Fermentation**

a. Use the yeast colonies that gave you the highest yield of the MIDEAS protein in step 10 and transfer them to a large fermentation reactor with a histidine and glycerol rich liquid mixture. Glycerol will be the initial carbon source of the yeast cells.

b. Grow the batch culture of yeast cells until all the glycerol is completely consumed.
c. Then add a continuous feed of glycerol into the fermentation reactor to create limiting conditions for yeast cells to access a carbon source and increase the biomass in the reactor.
d. When ready, induce the production of the protein by adding in methanol to the fermentation reactor.
e. Once the protein is sufficiently produced, take the supernatant and send through a filter to remove the cells and cell debris.
f. Send the supernatant through an Immobilized Metalion Affinity Chromatography (IMAC).
g. Add in Enterokinase (EK) to remove the HisTag by cleaving the protein at the EK protease site.
h. Send the mixture through the subtractive IMAC column to retrieve the purified MIDEAS protein.

**Protein Sequence of the MIDEAS Protein**

MNLQAQPKAQNKRKRCLFGGQEPAPKEQPPPLQPPQQSIRVKEEQYLGHEGPGGAVSTS
QPVELPPPSSLALLNSVVYGPERTSAAMLSQQVASVKWPNSVMAPGRGPERGGGGGVS
DSSWQQQPGQPPPHSTWNCHSLSLYSATKGSPHPGVGVPTYYNHPEALKREKAGGPQL
DRYVRPMMPQKVQLEVGRPQAPLNSFHAAKKPPNQSLPLQPFQLAFGHQVNRQVFRQG
PPPPNPVAAFPPQKQQQQQQPQQQQQQQAALPQMPLFENFYSMPQQPSQQPQDFGLQ
PAGPLGQSHLAHHSMAPYPFPPNPDMNPELRKALLQDSAPQPALPQVQIPFPRRSRRLSK
EGILPPSALDGAGTQPGQEATGNLFLHHWPLQQPPPGSLGQPHPEALGFPLELRESQLLP
DGERLAPNGREREAPAMGSEEGMRAVSTGDCGQVLRGGVIQSTRRRRRASQEANLLTL
AQKAVELASLQNAKDGSGSEEKRKSVLASTTKCGVEFSEPSLATKRAREDSGMVPLIIPV
SVPVRTVDPTEAAQAGGLDEDGKGPEQNPAEHKPSVIVTRRRSTRIPGTDAQAQAEDM
NVKLEGEPSVRKPKQRPRPEPLIIPTKAGTFIAPPVYSNITPYQSHLRSPVRLADHPSERSF
ELPPYTPPPILSPVREGSGLYFNAIISTSTIPAPPPITPKSAHRTLLRTNSAEVTPPVLSVMGE
ATPVSIEPRINVGSRFQAEIPLMRDRALAAADPHKADLVWQPWEDLESSREKQRQVEDL
LTAACSSIFPGAGTNQELALHCLHESRGDILETLNKLLLKKPLRPHNHPLATYHYTGSDQ
WKMAERKLFNKGIAIYKKDFFLVQKLIQTKTVAQCVEFYYTYKKQVKIGRNGTLTFGD
VDTSDEKSAQEEVEVDIKTSQKFPRVPLPRRESPSEERLEPKREVKEPRKEGEEEVPEIQE
KEEQEEGRERSRRAAAVKATQTLQANESASDILILRSHESNAPGSAGGQASEKPREGTG
KSRRALPFSEKKKKTETFSKTQNQENTFPCKKCGR

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